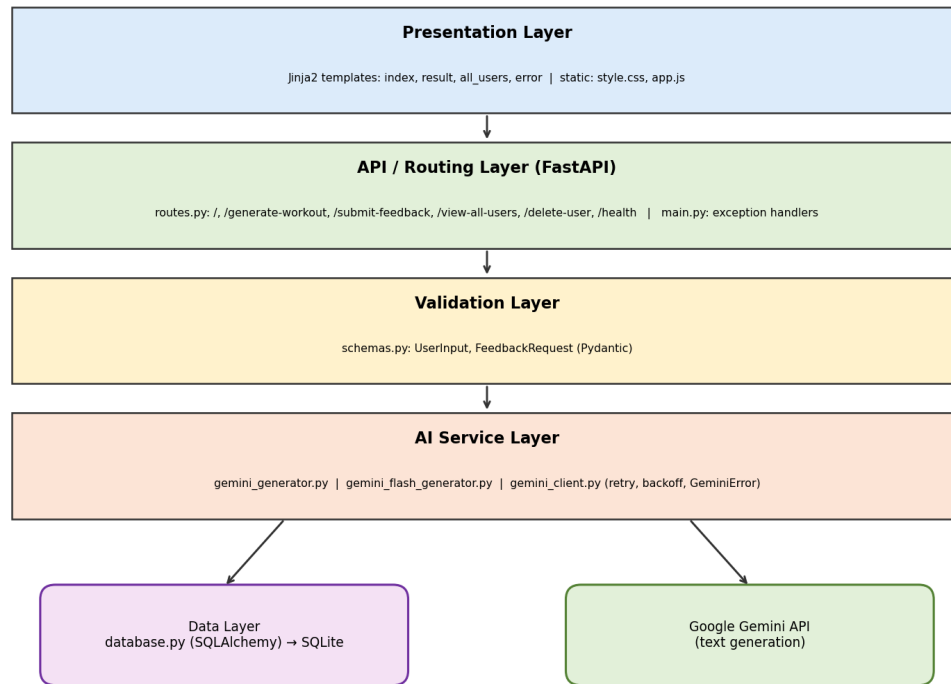


Solution Architecture

Project Design Phase · AI/ML and GenAI Track

FitBuddy is a layered monolith: a browser talks to a FastAPI application, which validates input, calls Gemini, and stores data in SQLite.

FitBuddy – Solution Architecture (Browser → Layers → Data / Gemini)

Layer	Components	Responsibility
Presentation	templates/*.html, static/style.css, static/app.js	Forms, result and admin pages, loading overlay
API / routing	app/main.py, app/routes.py	HTTP routes and exception handlers
Validation	app/schemas.py	Pydantic models with constraints
AI services	gemini_client.py, gemini_generator.py, gemini_flash_generator.py	Prompts, Gemini calls, retry/backoff, GeminiError
Data access	app/database.py	SQLAlchemy User model, session handling, CRUD
External	Google Gemini API; SQLite file	Text generation; storage

Request flow: Browser → FastAPI route → Pydantic validation → Gemini service (with retry) → Database → Jinja2 template → Browser.